Gal Arnon

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Appointments

Postdoctoral Researcher **2025-Present**
Bocconi University, supervised by Prof. Alon Rosen
Milan, Italy

Research Fellow **2025**
The Simons Institute for the Theory of Computing, UC Berkeley
Berkeley, USA

Education

Ph.D. Computer Science **2020-2025**
Weizmann Institute of Science
Advised By: Prof. Moni Naor and Dr. Eylon Yogev
Thesis: New Advancements in Interactive Oracle Proofs: Theory, Practice, and Limitations

M.Sc. Computer Science **2017–2020**
Weizmann Institute of Science
Advised By: Prof. Guy N. Rothblum
Thesis: On Prover-Efficient Public-Coin Emulation of Interactive Proofs

B.Sc. Electrical Engineering and Computer Science **2013–2017**
Tel Aviv University
Magna Cum Laude

Research Interests

Foundations of cryptography, computational complexity and theory of computation, probabilistic proof systems in both theory and practice.

Awards

- **Esther Hellinger Memorial Prize for academic excellence**. *Awarded in 2024 by the Weizmann Institute.*
- **Best Paper Award** for "STIR: Reed–Solomon Proximity Testing with Fewer Queries". *At Advances in Cryptology, the 44th Annual International Cryptology Conference (CRYPTO 2024).*

Publications

14. **Interactive Proofs for Batch Polynomial Evaluation**. Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, and Eylon Yogev. *(Manuscript)*
13. **Pairing-Based SNARGs with Two Group Elements**. Gal Arnon, Jesko Dujmovic, and Eylon Yogev. *(Manuscript)*
12. **Designated-Verifier SNARGs with One Group Element**. Gal Arnon, Jesko Dujmovic, and Yuval Ishai. *In proceedings of Advances in Cryptology, the 45th Annual International Cryptology Conference (CRYPTO 2025).*
11. **Towards a White-Box Secure Fiat-Shamir Transformation**. Gal Arnon and Eylon Yogev. *In proceedings of Advances in Cryptology, the 45th Annual International Cryptology Conference (CRYPTO 2025).* Additionally appeared in ZKProof 7. Work discussed in **Quanta Magazine** (`https://www.quantamagazine.org/computer-scientists-figure-out-how-to-prove-lies-20250709/`)
10. **Instance Compression, Revisited**. Gal Arnon, Shany Ben-David, and Eylon Yogev. *In proceedings of the 44th Annual International Conference on Theory and Application of Cryptographic Techniques (EUROCRYPT 2025).*

9. **WHIR: Reed–Solomon Proximity Testing with Super-Fast Verification**. Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, and Eylon Yogev. *In proceedings of the 44th Annual International Conference on Theory and Application of Cryptographic Techniques (EUROCRYPT 2025). Additionally appeared in ZKSummit 12 and ZKProof 7.*

8. **Hamming Weight Proofs of Proximity with One-Sided Error**. Gal Arnon, Shany Ben-David, and Eylon Yogev. *In proceedings of the 22nd Theory of Cryptography Conference (TCC 2024).*

7. **STIR: Reed–Solomon Proximity Testing with Fewer Queries**. Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, and Eylon Yogev. *In proceedings of Advances in Cryptology, the 44th Annual International Cryptology Conference (CRYPTO 2024)*, Part X, pp. 380–413. **Best Paper Award**. *Additionally appeared in ZKSummit 11 and ZKProof 6, and discussed in interview with me on ZK Podcast (https://zeroknowledge.fm/podcast/321/)*

6. **IOPs with Inverse Polynomial Soundness Error**. Gal Arnon, Alessandro Chiesa, and Eylon Yogev. *In proceedings of the 64th IEEE Annual Symposium on Foundations of Computer Science (FOCS 2023), pp. 752–761.*

5. **A Toolbox for Barriers on Interactive Oracle Proofs**. Gal Arnon, Amey Bhangale, Alessandro Chiesa, and Eylon Yogev. *In proceedings of the 20th Theory of Cryptography Conference (TCC 2022), pp. 447–466*

4. **Hardness of Approximation for Stochastic Problems via Interactive Oracle Proofs**. Gal Arnon, Alessandro Chiesa, and Eylon Yogev. *In proceedings of the 37th Annual IEEE Conference on Computational Complexity (CCC 2022), pp. 24:1–24:16.*

3. **Min-Entropic Optimality**. Gal Arnon and Tomer Grossman. *(Manuscript.)*

2. **A PCP Theorem for Interactive Proofs and Applications**. Gal Arnon, Alessandro Chiesa, and Eylon Yogev. *In proceedings of the 41st Annual International Conference on Theory and Application of Cryptographic Techniques (EUROCRYPT 2022), pp. 64–94.*

1. **On Prover-Efficient Public-Coin Emulation of Interactive Proofs**. Gal Arnon and Guy N. Rothblum. *In proceedings of the 2nd Conference on Information-Theoretic Cryptography (ITC 2021), volume 199 of LIPIcs, pp. 3:1–3:15.*

Invited Talks And Interviews

- Towards a White-Box Secure Fiat-Shamir Transformation. *Proofs Workshop at "Cryptography, 10 Years Later" Semester, Simons Institute, Berkeley, USA. July 2025.*
- IOPs with Inverse Polynomial Soundness Error. *ITC 2024 Highlights Track, Stanford University, USA. August 2024.*
- Interview about the STIR protocol for ZK Podcast along with Giacomo Fenzi. *Virtual, April 2024.* (https://zeroknowledge.fm/podcast/321/)
- How to convince someone who's barely listening (even to themselves). *At Efficient Probabilistic Proofs Workshop, Bertinoro, Italy. July 2022.*

Long-Term Visits

- Proofs, Consensus, and Decentralizing Society Semester at the Simons Institute, UC Berkeley. August-October 2019.

Service

Proximity Prize: Committee member/judge for "Proximity Prize" bounty put up by the Ethereum Foundation to prove or disprove Reed-Solomon proximity gaps conjectures.

Workshop Organization: Lattices Meet Hashes: Recent Advances in Post-Quantum Zero-Knowledge Proofs. *Postdoctoral Workshop at the Bernoulli Center, EPFL, Lausanne, Switzerland. Organized together with Ngoc Khanh Nguyen. May 2023.*

Conference Reviews: CCC (2024), CRYPTO (2019, 2022, 2023, 2024, 2025, 2026), EUROCRYPT (2023, 2026), FOCS (2025), ITC (2025), ITCS (2022, 2026), SICOMP (2026), SODA (2024), STOC (2025), TCC (2021, 2023, 2025)

Selected Talks

- Various Talks about Fiat-Shamir Attacks and Mitigations
  - Proofs Workshop at "Cryptography, 10 Years Later" Semester, Simons Institute, Berkeley, USA. July 2025.
  - Proofs Reading Group at "Cryptography, 10 Years Later" Semester, Simons Institute, Berkeley, USA. June 2025.
  - NTT Cryptography Seminar, (Virtual). April 2025.
- Instance Compression, Revisited
  - Foundations Reading Group at "Cryptography, 10 Years Later" Semester, Simons Institute, Berkeley, USA. July 2025.
- STIR: Reed–Solomon Proximity Testing with Fewer Queries
  - *CRYPTO 2024, Santa Barbara, USA. August 2024.*
  - *Interuniversity TCS Student Seminar, Tel Aviv University, Tel Aviv, Israel. May 2024.*
  - *Theory Lunch at the Weizmann Institute of Science, Rehovot, Israel. May 2024.*
  - *HUJI TCS Seminar, Jerusalem, Israel. May 2024.*
  - *StarkWare Industries, Netanya, Israel. April 2024.*
  - *ZKSummit 11, Athens, Greece. April 2024.*
- IOPs with Inverse Polynomial Soundness Error
  - *ITC 2024 Highlights Track, Stanford University, USA. August 2024.*
  - *Technion TCS Seminar, Haifa, Israel. February 2024.*
  - *ZK Study Club, Virtual. October 2023.*
  - *StarkWare Industries, Netanya, Israel. September 2023.*
  - *Interuniversity TCS Student Seminar, Tel Aviv University, Tel Aviv, Israel. July 2023.*
  - *IST Austria TCS Seminar, Vienna, Austria. June 2023.*
- A Toolbox for Barriers on Interactive Oracle Proofs
  - *TCC 2022, Chicago, USA. November 2022.*
- How To Be Convinced While Barely Listening (Even to Yourself)
  - *Theory Lunch at the Weizmann Institute of Science, Rehovot, Israel. May 2025.*
  - *EPFL CS Theory Reading Group, Lausanne, Switzerland. May 2023.*
  - *Efficient Probabilistic Proofs Workshop, Bertinoro, Italy. July 2022. (Talk given under alternate title.)*
- Hardness of Approximation for Stochastic Problems via Interactive Oracle Proofs
  - *CCC 2022, Philadelphia, USA. July 2022. (Talk given virtually.)*
- A PCP Theorem for Interactive Proofs and Applications
  - *EUROCRYPT 2022, Trondheim, Norway. May-June 2022.*
  - *Theory Lunch at the Weizmann Institute of Science, Rehovot, Israel. July 2021.*
- On Prover-Efficient Public-Coin Emulation of Interactive Proofs
  - *ITC 2021, Virtual. July 2021.*
  - *"Proofs, Consensus, and Decentralizing Society" Program Seminar at Simons Institute, Berkeley, USA. October 2019.*

Teaching

Teaching Assistant:

- Foundations and Frontiers of Probabilistic Proofs. *MSRI (SLMath) summer graduate school. Zurich, Switzerland. July 2023.*

- Foundations and Frontiers of Probabilistic Proofs. *MSRI summer graduate school. Held virtually. July-August 2021.*

Instructor: Mini-Course on Zero-Knowledge Proofs. *Amos de-Shalit Summer School, Weizmann Institute of Science. September 2018.*